Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2023 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

28 adaptive question sections   103 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 103 marks.

Contents

Question 01

4MA1-1H Q1

MODULAR UNIT 2

3 marks

1 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2:5:8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

....

WORKING SPACE

Question 02
4MA1-1H Q2

MODULAR UNIT 2
3 marks

2 The table gives information about the number of minutes that Abby spent walking each day in September.

Number of minutes (M)	**Frequency**
$0 < M \leqslant 30$	5
$30 < M \leqslant 60$	6
$60 < M \leqslant 90$	8
$90 < M \leqslant 120$	9
$120 < M \leqslant 150$	2

Work out an estimate for the total number of minutes that Abby spent walking in September.

.... minutes

WORKING SPACE

Question 03

4MA1-1H Q3

MODULAR UNIT 2

4 marks

3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.
Give your answer correct to 3 significant figures.

....%

WORKING SPACE

Question 04

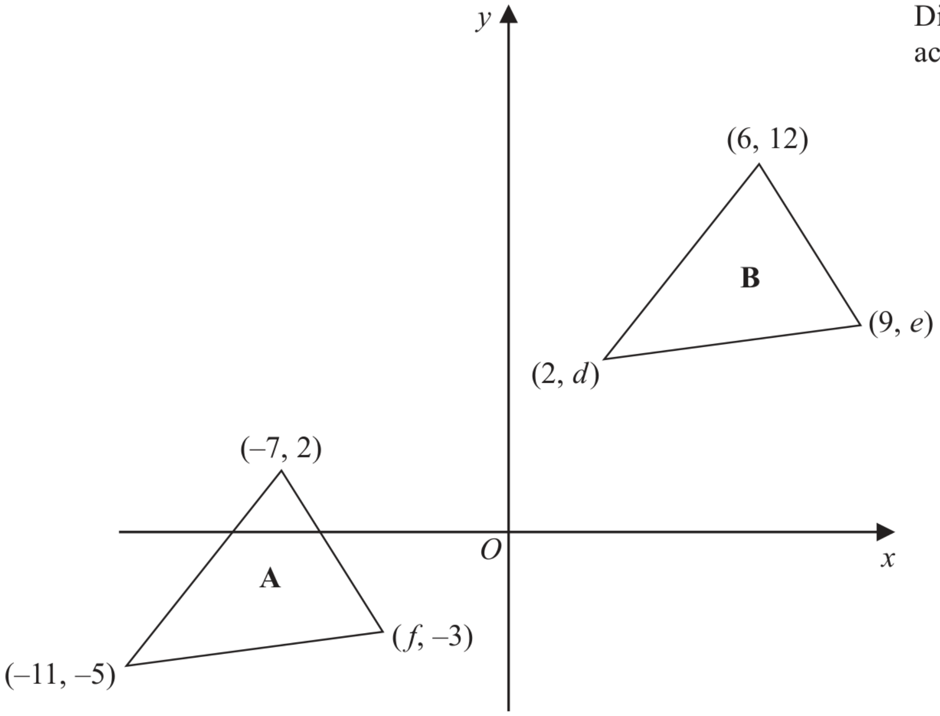
4MA1-1H Q4

MODULAR UNIT 2

3 marks

4 The diagram shows a sketch of triangle **A** and triangle **B** on a coordinate grid.

Diagram **NOT** accurately drawn



(a) Given that triangle **A** has been translated to give triangle **B**, work out the value of d, the value of e and the value of f

$d =$ ....

$e =$ ....

$f =$ ....

(3)

Working space for Question 04

Continue your method

UNIT 2
3 marks

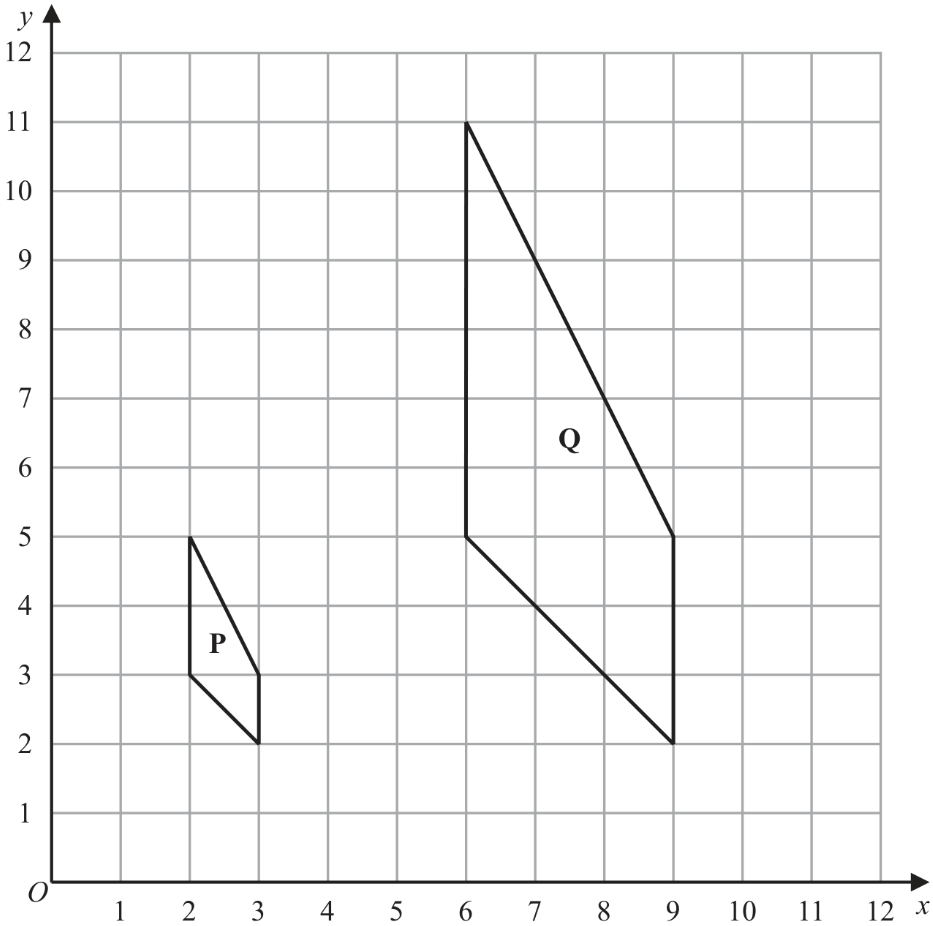
WORKING SPACE

Question 05

4MA1-1H Q4

MODULAR UNIT 2
5 marks

The diagram shows shape **P** and shape **Q** drawn on a grid.



(b) Describe fully the single transformation that maps shape **P** onto shape **Q**

....

....

(3)

(c) On the grid above, rotate shape **P** 90° clockwise about (3, 5)
Label your shape **R**

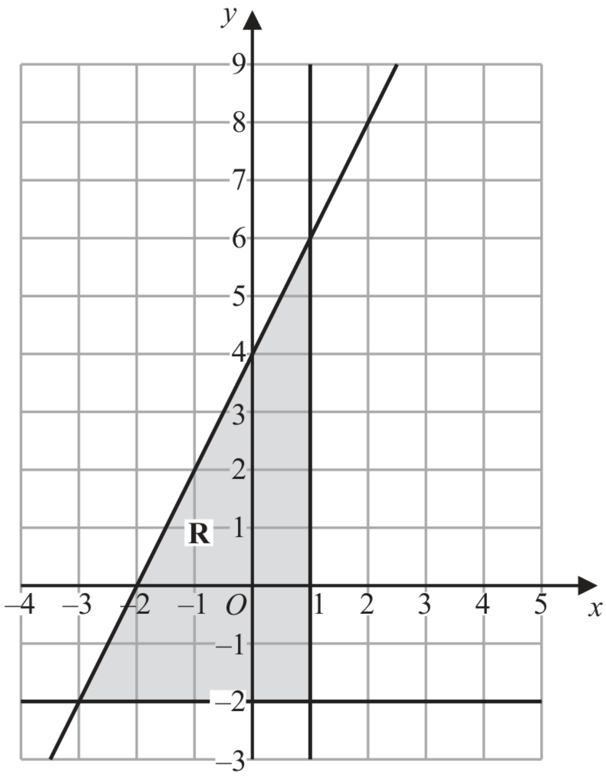
(2)

Question 06

4MA1-1H Q7

MODULAR UNIT 2

4 marks



The region **R**, shown shaded in the diagram, is bounded by three straight lines.

Find the inequalities that define **R**

....

....

....

Working space for Question 06

Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 07

4MA1-1H Q8

MODULAR UNIT 2
2 marks

8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

....
(2)

WORKING SPACE

Question 08

4MA1-1H Q8

MODULAR UNIT 2
2 marks

$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$

(b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

....

(2)

WORKING SPACE

Question 09

4MA1-1H Q9

MODULAR UNIT 2

3 marks

9 Solve the simultaneous equations

$$2x + 9y = 14.5$$
$$7x + 3y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

Question 10

4MA1-1H Q10

MODULAR UNIT 2
2 marks

10 Here are the test marks of 15 students.

7 10 14 15 16 17 18 19 20 20 23 25 30 39 40

Find the interquartile range of these marks.

....

WORKING SPACE

Question 11
4MA1-1H Q11

MODULAR UNIT 2
6 marks

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\dfrac{dy}{dx}$

$\dfrac{dy}{dx} =$ ....

(2)

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

....

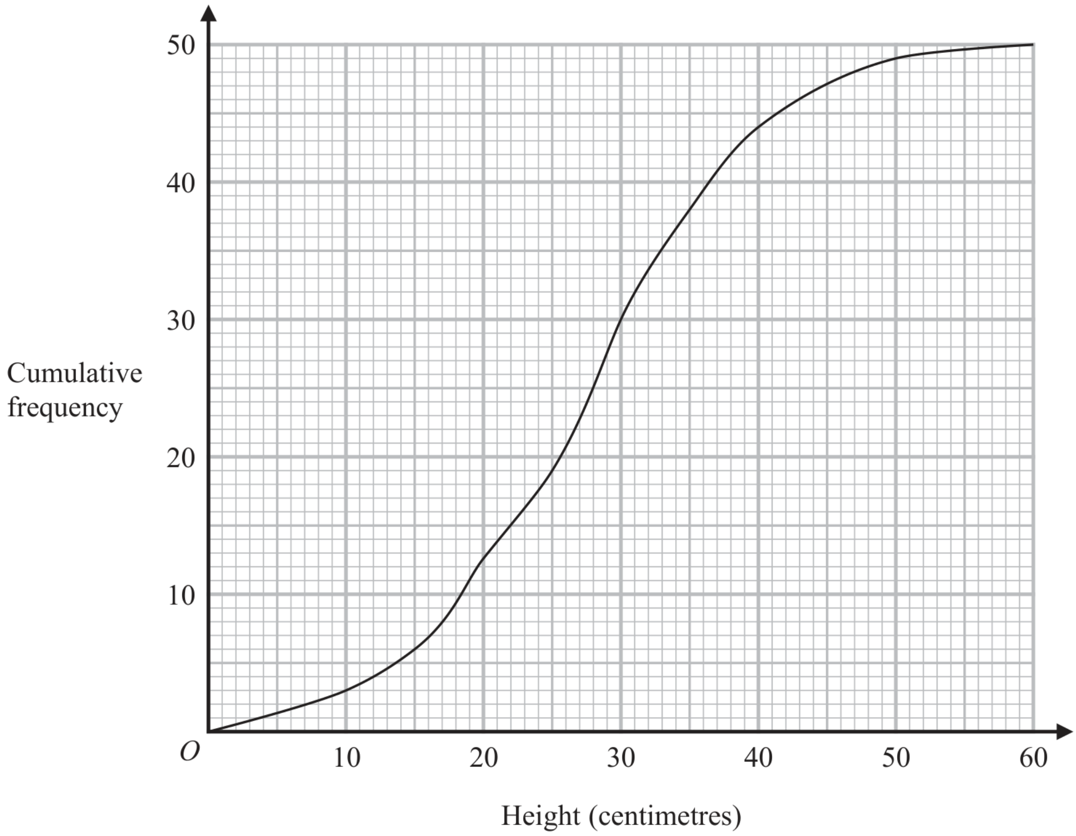
(4)

WORKING SPACE

Question 12
4MA1-1H Q12

MODULAR UNIT 2
4 marks

12 The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



(a) Use the graph to find an estimate for the median height of these plants.

.... centimetres
(1)

(b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leqslant 40$

....
(1)

(c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

....
(2)

Working space for Question 12

Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 13

4MA1-1H Q15

MODULAR UNIT 2
3 marks

(b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

....
(3)

WORKING SPACE

Question 14

4MA1-1H Q17

MODULAR UNIT 2
4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \qquad x \geqslant 0$$

(a) What value of x must be excluded from any domain of g?

....

(1)

(b) Solve $gh(x) = 1$

....

(3)

WORKING SPACE

Question 15

4MA1-1H Q19

MODULAR UNIT 2
5 marks

19

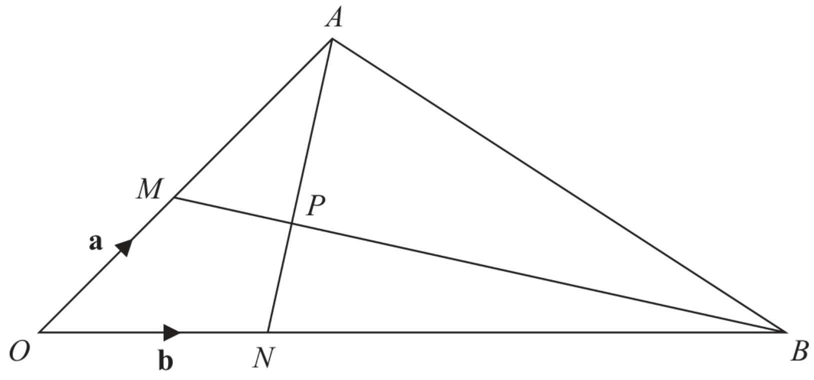


Diagram **NOT** accurately drawn

OMA, ONB, MPB and NPA are straight lines.

M is the midpoint of OA

$ON : NB = 1 : 5$

$\overrightarrow{OM} = \mathbf{a} \qquad \overrightarrow{ON} = \mathbf{b}$

(a) Find in terms of **a** and **b** the vector $\overrightarrow{AN}$

....
(1)

(b) Use a vector method to find the ratio $AP : PN$

$AP : PN =$ ....
(4)

Working space for Question 15

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 16

4MA1-1H Q20

MODULAR UNIT 2
6 marks

20 The sum of the first 80 terms of an arithmetic series, S, is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X
Show your working clearly.

$X =$ ....

WORKING SPACE

Question 17

4MA1-1H Q21

MODULAR UNIT 2
2 marks

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.
The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.......................... , ...)
(1)

(b) $y = f(3x)$

(.......................... , ...)
(1)

WORKING SPACE

Question 18

4MA1-1H Q23

MODULAR UNIT 2
5 marks

23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.

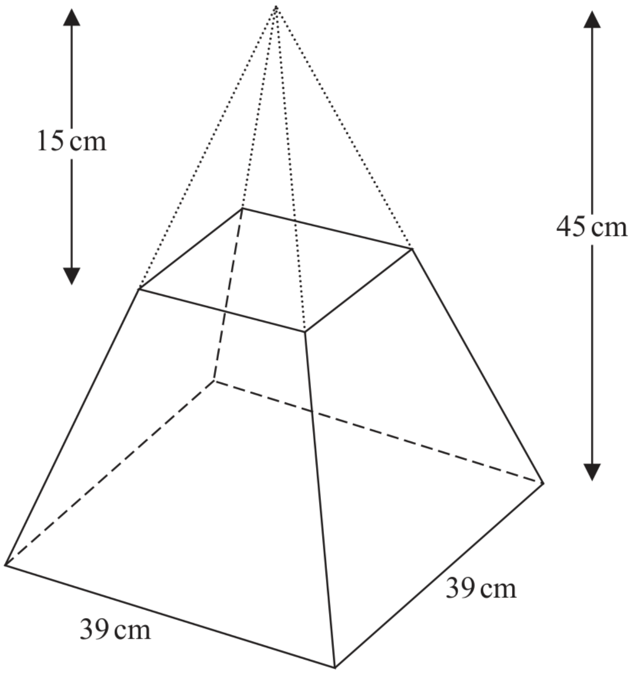


Diagram **NOT** accurately drawn

The height of the small pyramid is 15 cm.
The height of the large pyramid is 45 cm.
The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.
Give your answer correct to the nearest whole number.

.... cm^2

Working space for Question 18

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 19

4MA1-2H Q5

MODULAR UNIT 2

4 marks

5 Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

....

WORKING SPACE

Question 20

4MA1-2H Q6

MODULAR UNIT 2
2 marks

6 (a) Write 76 000 000 in standard form.

....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

....
(1)

WORKING SPACE

Question 21
4MA1-2H Q7

MODULAR UNIT 2
3 marks

7

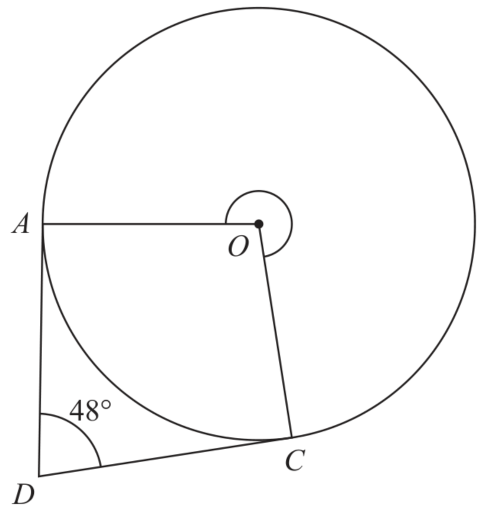


Diagram **NOT** accurately drawn

A and C are points on a circle, centre O
DA is the tangent to the circle at A and DC is the tangent to the circle at C

Angle $ADC = 48°$

Work out the size of reflex angle AOC

.... °

Working space for Question 21

Continue your method

UNIT 2
3 marks

WORKING SPACE

Question 22

4MA1-2H Q8

MODULAR UNIT 2
3 marks

8 Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$...

WORKING SPACE

Question 23

4MA1-2H Q10

MODULAR UNIT 2
4 marks

10 Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17
After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.
After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

....

WORKING SPACE

Question 24

4MA1-2H Q11

MODULAR UNIT 2

3 marks

11 Here is a 9-sided regular polygon *ABCDEFGHJ*, with centre *O*

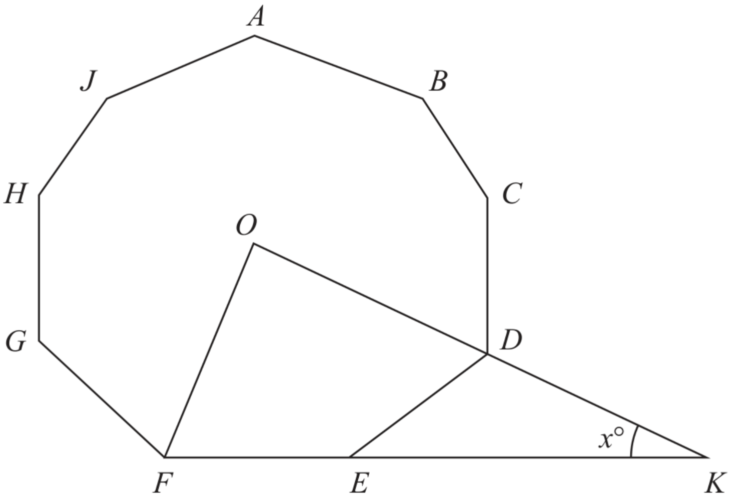


Diagram **NOT** accurately drawn

ODK and *FEK* are straight lines.

Work out the value of x

$x =$ ....

Working space for Question 24

Continue your method

UNIT 2
3 marks

WORKING SPACE

Question 25
4MA1-2H Q14

MODULAR UNIT 2
4 marks

14

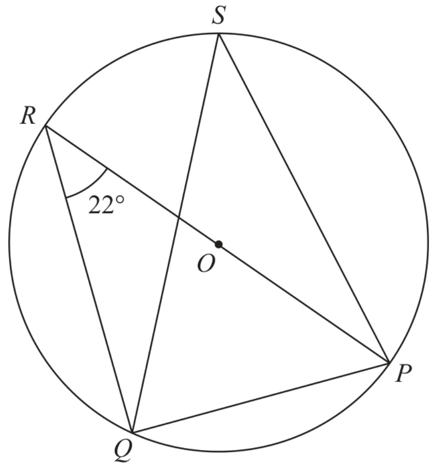


Diagram **NOT** accurately drawn

P, Q, R and S are points on a circle, centre O
ROP is a diameter of the circle.
Angle $PRQ = 22°$

(a) (i) Find the size of angle RQP

.... °
(1)

(ii) Give a reason for your answer.

....

....
(1)

(b) (i) Find the size of angle PSQ

.... °
(1)

(ii) Give a reason for your answer.

....

....
(1)

Working space for Question 25

Continue your method

UNIT 2
4 marks

WORKING SPACE

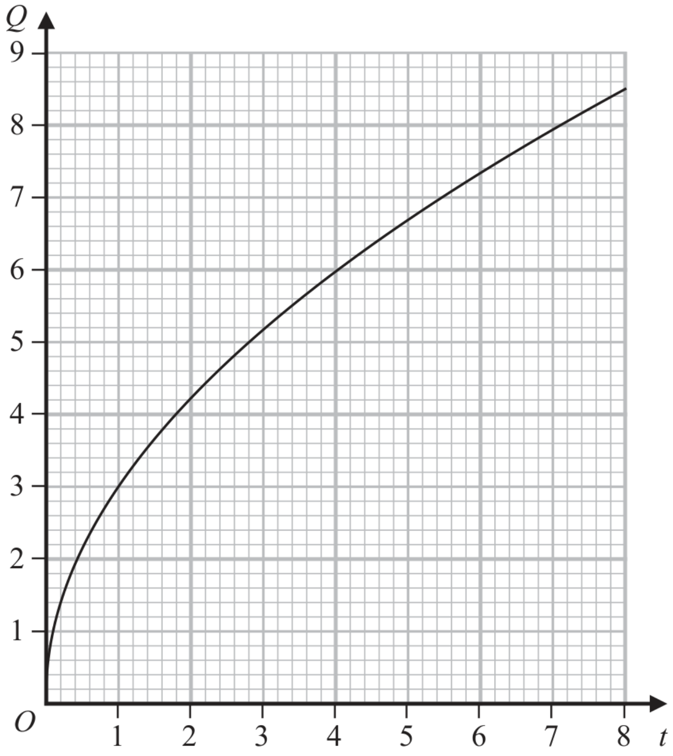
Question 26

4MA1-2H Q16

MODULAR UNIT 2
5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

....

(3)

Q is increased by 20%

(b) Find the percentage increase in t

.... %

(2)

Working space for Question 26

Continue your method

UNIT 2
5 marks

WORKING SPACE

Question 27

4MA1-2H Q17

MODULAR UNIT 2

4 marks

(b) Make e the subject of $w = \sqrt{\dfrac{e+g}{ef-d}}$

....

(4)

WORKING SPACE

Question 28

4MA1-2H Q21

MODULAR UNIT 2

5 marks

21 Solve the simultaneous equations

$$2x^2 + 3y^2 = 11$$
$$x = 3y - 1$$

Show clear algebraic working.

....

WORKING SPACE